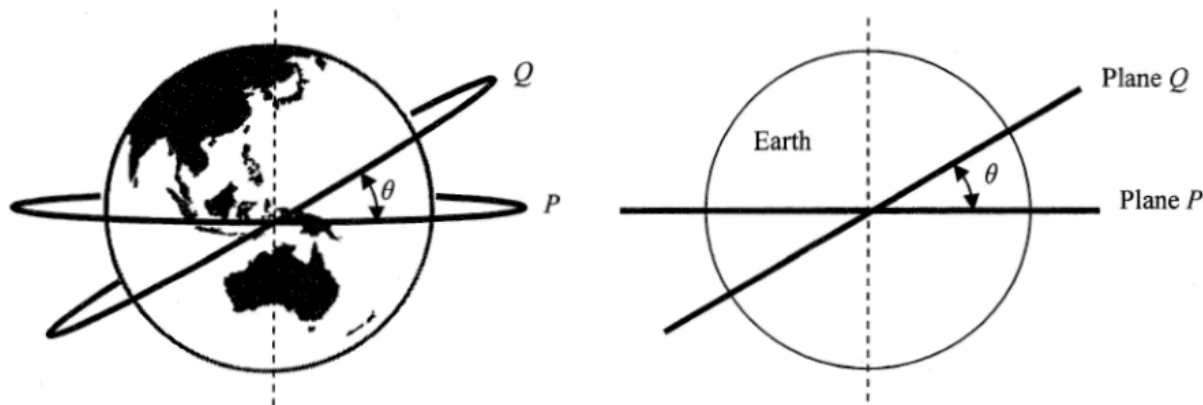


- *14. Two satellites move in circular orbits of the same radius R around the Earth (mass M). The orbits are in two different planes P and Q as shown. Plane P coincides with the Earth's equator while plane Q is inclined to the equator at θ . Which statement is **INCORRECT**?



- A. The speed of satellite P is $\sqrt{\frac{GM}{R}}$.
- B. The centripetal force acting on satellite Q is pointing along the plane Q .
- C. The acceleration of both satellites is the same in magnitude.
- D. The period of satellite Q is longer than that of satellite P .

- *15. It is known that the mass of Mars is about $\frac{1}{10}$ of that of the Earth while its radius is about $\frac{1}{2}$ of the Earth's radius. In terms of the gravitational acceleration g on the Earth's surface, the approximate gravitational acceleration on the surface of Mars is

- A. $0.2 g$.
- B. $0.4 g$.
- C. $2.5 g$.
- D. $4 g$.

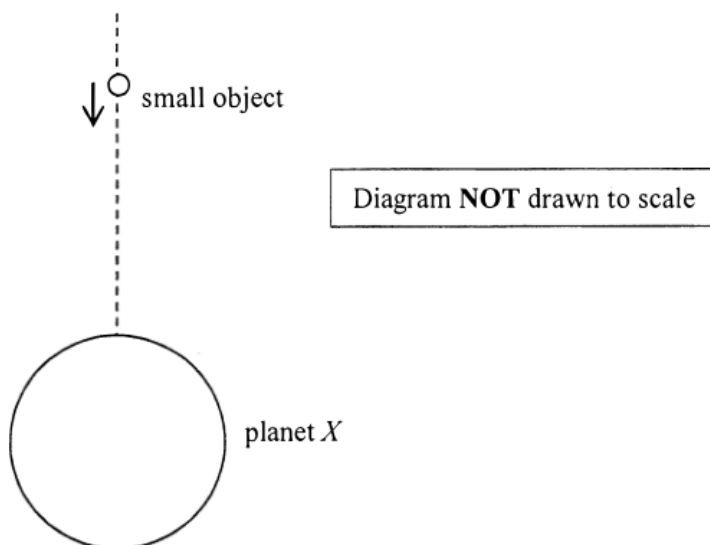
- *11. An astronaut inside a spacecraft moving in a circular orbit around the Earth is apparently weightless because
- A. the astronaut is too far from the Earth to feel the Earth's gravitational force.
 - B. the astronaut and the spacecraft are both moving with the same acceleration towards the Earth.
 - C. the Earth's gravitational force on the astronaut is balanced by the reaction force of the spacecraft's floor.
 - D. the Earth's gravitational force on the astronaut is balanced by the centripetal force.

- *12. An artificial satellite revolves in a circular orbit above the Earth's surface at a height equal to the radius of the Earth. Find the acceleration of the satellite in terms of the acceleration due to gravity g on the Earth's surface.
- A. $\frac{g}{8}$
 - B. $\frac{g}{4}$

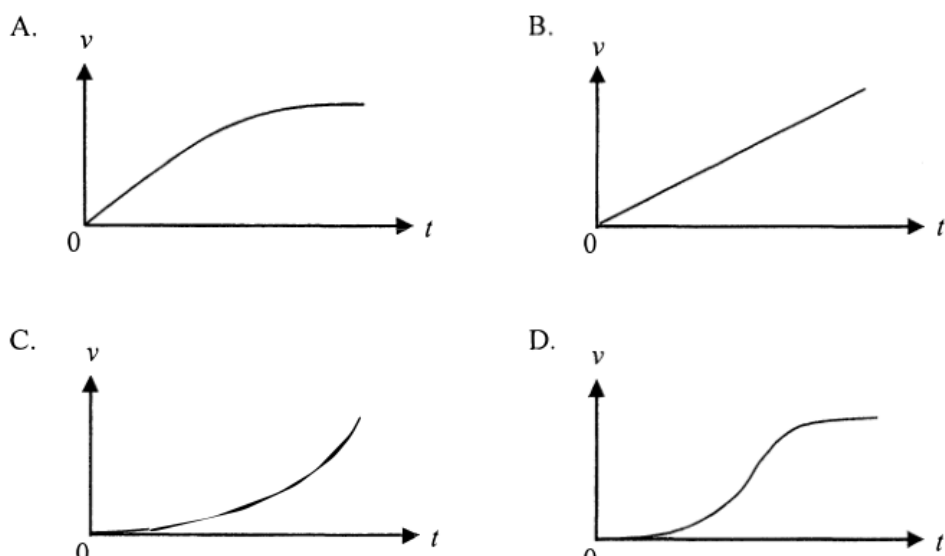
11. The gravitational force exerted on the Earth by the Sun is F_0 . The gravitational force exerted on the Sun by the Earth is
- A. equal to F_0 and in the same direction.
 - B. equal to F_0 and in the opposite direction.
 - C. much smaller than F_0 and in the same direction.
 - D. much smaller than F_0 and in the opposite direction.

- *14. A satellite orbits the Earth in a circular path of radius 7.2×10^6 m. What is the period of the satellite ?
Given : mass of the Earth = 6.0×10^{24} kg
- A. 1.4 hours
 - B. 1.7 hours
 - C. 1 day
 - D. Answer cannot be found as the mass of the satellite is not known.

- *13. A small object is released from rest at a point very far away from a planet X . The object then starts moving towards X . X does not have an atmosphere. Neglect the effect of other celestial bodies.



Which of the following graphs best shows the variation of the velocity v of the object with time t before it hits X ?



- *13. A satellite of mass m moves around a planet of mass M in a circular orbit of radius r . What does the angular velocity of the satellite depend on?

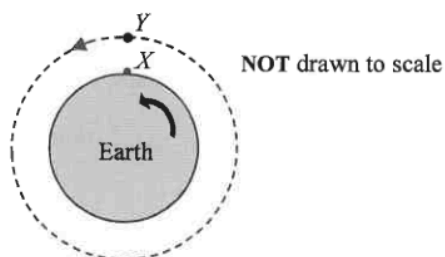
- (1) r
 (2) m
 (3) M

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

- *10. The diameter of Neptune is about 4 times that of the Earth and its mass is about 17 times that of the Earth. Estimate the acceleration due to gravity on Neptune's surface.
Given: acceleration due to gravity on Earth's surface $g = 9.81 \text{ m s}^{-2}$

- A. 2.3 m s^{-2}
- B. 9.2 m s^{-2}
- C. 10.4 m s^{-2}
- D. 41.7 m s^{-2}

- *14. In the figure, X is an object resting on the Earth's equator. Y is a geostationary satellite moving in a circular orbit above the equator such that it always appears stationary to an observer on Earth.



Which descriptions about the motion of X and Y are correct ?

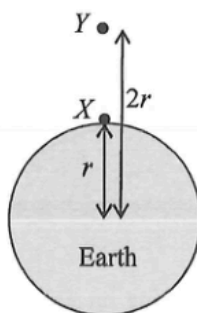
- (1) The motion of X and Y takes the same period.
- (2) X is moving with a slower speed.
- (3) X has a greater acceleration.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- *13. A satellite orbits the Earth in a circular orbit of radius r with a period of 24 hours. The period of another satellite which orbits the Earth in a circular orbit of radius $2r$ is

- A. less than 24 hours.
- B. 24 hours.
- C. between 24 hours and 48 hours.
- D. more than 48 hours.

*12.



Two identical objects located at point X and point Y are at distances r and $2r$ from the Earth's centre. The weight of the object at point X is W . What is the weight of the object at point Y ?

- A. $0.25W$
- B. $0.5W$
- C. $0.75W$
- D. W

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- *11. Satellites X and Y revolve around the Earth in circular orbits, in the same plane and in the same direction. The radii of their orbits are r and $4r$ respectively. The figure below shows an instant that X , Y and the Earth are on a straight line.

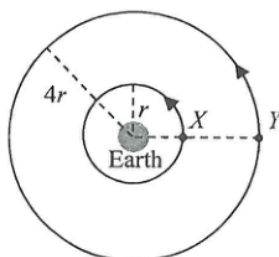
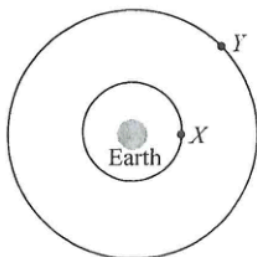


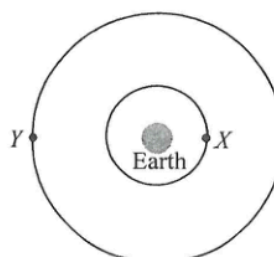
Diagram NOT drawn to scale

When X completes one cycle, which of the following diagrams best represents the position of Y ?

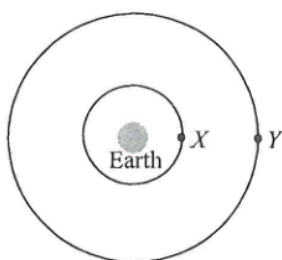
A.



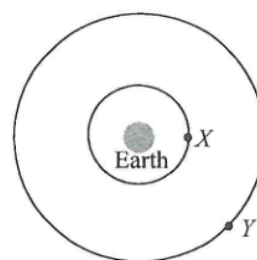
B.



C.



D.



- *12. A rocket on the Earth's surface is launched by an upward thrust F , producing an upward acceleration g , where g is the acceleration due to gravity near the Earth's surface, as shown in Figure (a). R is the radius of the Earth.

Diagram NOT drawn to scale

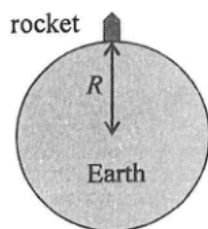


Figure (a)

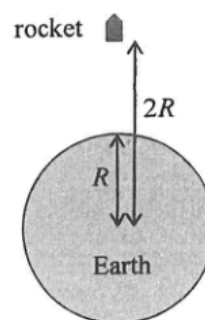


Figure (b)

Figure (b) shows an identical rocket at a distance $2R$ away from the Earth's centre. What is the upward thrust required for producing the same upward acceleration g ? Neglect air resistance.

- A. $\frac{F}{4}$
B. $\frac{F}{2}$
C. $\frac{5F}{8}$
D. $\frac{3F}{4}$